PHKI / HSLU / SIMON BALMER

JUST ONE MORE MOMENT

WIRING OVERVIEW

REV 04 / 28 OCT 2026

Connections grouped by component. Physical J8 pin positions: locator below.

01 / TOUCH DISPLAY 2

+5 V
GND
VIDEO + TOUCH
DSI

02 / ILLUMINATED ARCADE BUTTON

33 mm / WHITE LED

COM
NO
LED+
LED-

03 / SW01 / AI MODE

LOCAL AI /
CLOUD AI + CAM

SIGNAL
GND

04 / SW02 / LANGUAGE

EN / DE

SIGNAL
GND

RASPBERRY PI 5

8 GB / CONNECTION VIEW

P2 / 5V
P6 / GND
CAM/DISP / DSI
P9 / GND
P11 / GPIO17
P13 / GPIO27
P14 / GND
P15 / GPIO22
P20 / GND
P16 / GPIO23
P25 / GND
Pxx = physical header pin

J8 / GPIO

BCM numbering
for GPIOxx

P4 / 5V
P19 / GPIO10
P34 / GND
USB
USB-C

D1 / MIC 1N4001

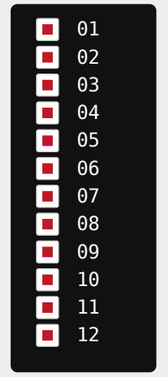
A K

RING -> STRIP

05 / WS2812B / 12 LEDs

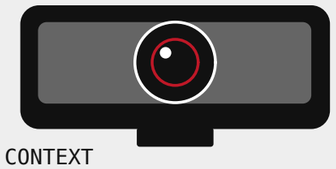
+V
DIN
GND

DIN -> PIXEL 01
-> ... -> 12



06 / LOGITECH C270

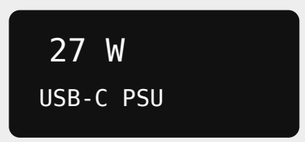
USB



USB / VISUAL CONTEXT

07 / OFFICIAL POWER SUPPLY

USB-C



J8 / PHYSICAL PIN LOCATOR

TOP VIEW / USB PORTS AT RIGHT

2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39

P1 = lower left. Red pins are used in my wiring.

5 V / +V GND GPIO / 3.3 V

Component and port positions in the main view are grouped for clarity.

WHY I USE D1

I lower the strip supply with D1 so it can recognize 3.3 V data more reliably. GPIO10 itself stays at 3.3 V.

Cathode ring -> strip +V. DIN -> GPIO10.

SW01 and SW02 are software inputs. SW01 does not switch the USB camera cable.

I disconnect power before changing wires.